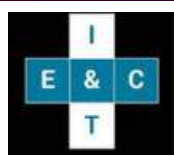


Post Graduation Diploma in **Professional MERN Stack** & **AI Development**

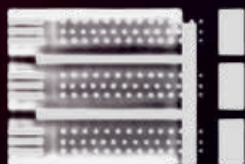
Duration: 8Months



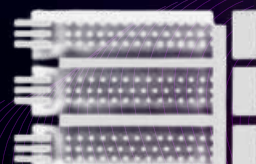
E&ICT Academy
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ABOUT PROGRAMS

This advanced certification program developed by the faculty at IIT Guwahati and industry experts at SkyySkill Academy, aims to bring you up to speed as well as deepen your understanding of Electric Vehicle Technology and other related concepts. The purpose of this course is to develop the next generation of EV system engineers who will advance the field of vehicle design.

ABOUT SKYYSKILL ACADEMY

Skyy Skill Academy is an Epitome of skill developing Institutions . It Provides wide range of courses which enables students to become empowered and employees through world class training. The practical & industrial approach has made SkyySkill the industry leader.

KEY HIGHLIGHTS



100% Live or Offline Session and
1:1 Mentorship



Placement Guaranteed Programs



Hands-on Practical & Live
Projects



Industry Recognized Courses &
Certificates



Industry Standard projects



24X7 doubt clearing assistance



On demand doubt clearing classes



Most affordable & Best in class



WHO CAN APPLY FOR THE COURSE ?

This diploma is designed for ambitious individuals ready to accelerate their careers in technology. Whether you're starting your journey or pivoting to development, our comprehensive curriculum accommodates diverse backgrounds.



Freshers & Engineering Graduates

Recent graduates seeking to launch their tech careers with industry-relevant skills and real-world project experience.



Systems & Network Professionals

IT professionals looking to transition into full-stack development and expand their technical skillset.



Computer Science Engineering

Engineers from traditional domains wanting to embrace software development and digital transformation.



Any Graduate

Motivated graduates from any discipline with a passion for technology and problem-solving.

LEARNING PATH

1

Foundation & Frontend Basics

5

MongoDB & Database Integration

2

Advanced JavaScript & React Basics

6

Advanced Topics & Deployment

3

Advanced React & State Management

7

Major Project & Internship

4

Backend Development with Node.js & Express



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ABOUT E&ICT ACADEMY, IIT GUWAHATI

E&ICT Academy is a joint initiative by the Ministry of Electronics & Information Technology (MeitY), Government of India and IIT Guwahati with the aim of training the forthcoming technical workforce to enhance employability and build capacity in the area of Electronics & Information and Communication Technology.

The Academy offer online Advanced Certifications in Data Science, AI & ML, Big Data, Cloud Computing, Web Development, VLSI Design, & Management-related courses and has trained 2500+ graduates & working professionals.

COLLABORATION WITH E&ICT ACADEMY IIT GUWAHATI

Collaboration With **E&ICT Academy IIT Guwahati** to improve technological education and research and innovation into the partnership with Skyy Skill Academy to jointly develop high-quality educational content curated by experts in the respective fields.





COURSE CONTENT

Month 1: Foundation & Frontend Basics

WEEK 1-2: HTML & CSS FUNDAMENTALS

- HTML5 semantic elements, forms, accessibility
- CSS3: Flexbox, Grid, positioning, animations
- Responsive design principles and mobile-first approach
- CSS preprocessors (Sass/SCSS basics)

WEEK 3-4: JAVASCRIPT FUNDAMENTALS

- Variables, data types, operators, control structures
- Functions, scope, closures, hoisting
- Arrays and objects, destructuring, spread/rest operators
- ES6+ features: arrow functions, template literals, modules
- Asynchronous JavaScript: callbacks, promises, async/await
- DOM manipulation and event handling
- Error handling and debugging

Month 2: Advanced JavaScript & React Basics

WEEK 1-2: ADVANCED JAVASCRIPT CONCEPTS

- Object-oriented programming in JavaScript
- Prototypes and inheritance
- this keyword and context binding
- Higher-order functions, map, filter, reduce
- Fetch API and working with REST APIs
- Local storage and session storage
- Introduction to Git and GitHub

WEEK 3-4: REACT FUNDAMENTALS

- Introduction to React and JSX
- Components (functional and class components)
- Props and state management
- Event handling in React
- Conditional rendering and lists
- React lifecycle methods and useEffect
- Introduction to React hooks (useState, useEffect)

Month 3: Advanced React & State Management

WEEK 1-2: ADVANCED REACT CONCEPTS

- Custom hooks and hook patterns
- useContext, useReducer, useMemo, useCallback
- React Router for navigation and routing
- Forms and controlled components (React Hook Form or Formik)
- Error boundaries and error handling
- Code splitting and lazy loading
- Performance optimization techniques

WEEK 3-4: STATE MANAGEMENT & TESTING

- Introduction to Redux or Context API with useReducer
- Redux Toolkit for modern Redux development
- Redux middleware (Thunk or Saga)
- React testing basics with Jest and React Testing Library
- Component testing strategies
- Integration with REST APIs

Month 4: Backend Development with Node.js & Express

WEEK 1-2: NODE.JS FUNDAMENTALS

- Introduction to Node.js and npm
- Node.js modules and module system
- File system operations
- Event loop and asynchronous programming
- Introduction to Express.js
- Routing and middleware in Express
- Request/response handling
- Error handling middleware



COURSE CONTENT

WEEK 2-3: BUILDING REST APIS

- RESTful API design principles
- CRUD operations with Express
- Request validation (express-validator or Joi)
- Environment variables and configuration
- Authentication concepts (JWT, sessions)
- Password hashing with bcrypt
- Authorization and role-based access control
- API security best practices (helmet, cors, rate limiting)

WEEK 4: ADVANCED BACKEND CONCEPTS

- File uploads with Multer
- Email integration (NodeMailer)
- Logging and monitoring
- API documentation with Swagger
- Testing APIs with Postman and Jest/Supertest

Month 5: MongoDB & Database Integration

WEEK 1-2: MONGODB FUNDAMENTALS

- Introduction to NoSQL and MongoDB
- MongoDB installation and setup
- CRUD operations in MongoDB
- MongoDB data modeling and schema design
- Mongoose ODM for Node.js
- Mongoose schemas, models, and validation
- Relationships in MongoDB (embedding vs referencing)
- Querying and aggregation

WEEK 3-4: FULL-STACK INTEGRATION

- Connecting React frontend with Express backend
- Building a complete CRUD application
- Handling authentication flow (login, signup, logout)
- Protected routes on frontend and backend
- Image upload and storage
- Pagination, sorting, and filtering
- Search functionality implementation
- Real-time features with Socket.io (optional)

Month 6: Advanced Topics & Deployment

WEEK 1-2: ADVANCED FULL-STACK FEATURES

- Third-party API integrations (payment gateways like Stripe)
- Social authentication (Google, Facebook OAuth)
- Advanced state management patterns
- Caching strategies (Redis basics)
- Background jobs and task queues
- Performance optimization (frontend and backend)
- SEO basics for React applications

WEEK 3-4: DEVOPS & DEPLOYMENT

- Version control with Git (branching, merging, pull requests)
- Introduction to CI/CD concepts
- Deploying React applications (Vercel, Netlify)
- Deploying Node.js applications (Heroku, Railway, Render)
- MongoDB Atlas for cloud database
- Environment management for different stages
- Domain configuration and SSL certificates
- Monitoring and error tracking (Sentry basics)
- Docker basics (containerization)

COURSE CONTENT

Month 7-8: Internship Period

WEEK 1-2: ONBOARDING & SETUP

- Company codebase familiarization
- Development environment setup
- Understanding existing architecture
- Code review process and standards
- Agile/Scrum methodology introduction
- Assigned mentor and initial tasks

WEEK 3-4: MINOR FEATURES & BUG FIXES

- Working on small feature implementations
- Bug fixing and testing
- Code reviews and feedback incorporation
- Documentation writing
- Learning company-specific tools and workflows

WEEK 5-6: MAJOR FEATURE DEVELOPMENT

- Assigned to larger features or modules
- Collaborative work with team members
- Database design and API development
- Frontend component development
- Writing unit and integration tests
- Participating in sprint planning and standups

WEEK 7-8: PROJECT COMPLETION & PRESENTATION

- Final project or feature completion
- Performance optimization and refactoring
- Documentation and knowledge transfer
- Final presentation to the team
- Feedback and performance review
- Portfolio project completion





COURSE CONTENT

PROMPT ENGINEERING & GENERATIVE AI FOR FULL STACK DEVELOPERS

Module 1: Prompt Engineering & Generative AI Basics

- What is Prompt Engineering?
- Overview of Generative AI (ChatGPT, Claude, Bard, Gemini)
- Developer's view: AI vs Classical Programming
- Setting up your AI Toolkit (OpenAI Playground, Postman, IDEs)

Module 2: AI for Frontend

Development

- Generate HTML, CSS, JS using AI
- Turn UI wireframes into responsive code
- Use AI tools for UI design (e.g., Figma AI, Uizard)
- Prompting for animations, layout, and content generation

Module 3: AI for Backend

Development

- Auto-generate REST APIs and database schemas
- Debugging and code review with AI tools
- Integrating OpenAI/Hugging Face APIs with Node.js
- AI-generated test cases and backend logic suggestions (auth, middleware)

Module 4: Real-World AI Use Cases for Fullstack Devs

- Build AI-powered chatbots using ChatGPT API
- Auto-generate documentation with AI + Swagger/OpenAPI
- Implement AI-powered search (semantic search, vector DBs)
- Use tools like GitHub Co-pilot, Tabnine for daily dev tasks

Module 5: Capstone Project

- Build an AI-enhanced full stack app (e.g., AI blog writer)
- Integrate an AI Chatbots into an existing React/Node.js app
- Create a prompt-based frontend code generator
- Includes: Peer reviews + Mentor feedback



PROGRAM COMPLETION CERTIFICATE



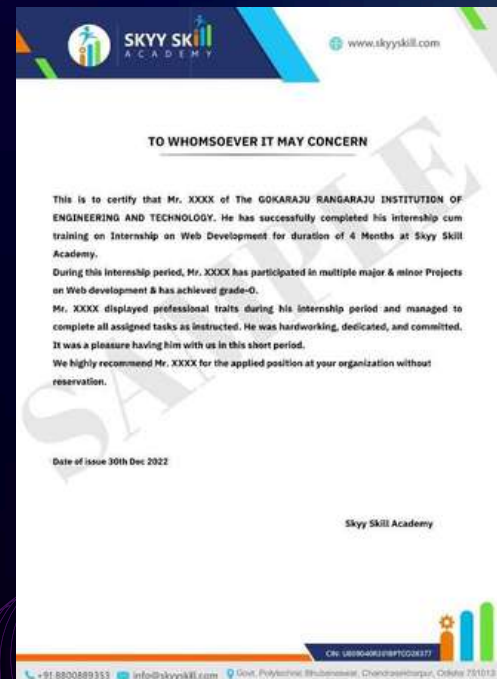
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